



The connection between Bayesian networks and adverse outcome pathway (AOP) networks and how to use it for predicting drug toxicity

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There is significant interest in combining adverse outcome pathways (AOPs) with Bayesian networks (BNs) because of their shared representation using directed acyclic graphs (DAGs). However, it has not been verified empirically whether AOP networks are mathematically congruent with BNs. Furthermore, important properties for BNs, such as Markov blankets, have not been emphasized, which is a missed opportunity for simplifying and optimizing the model. Here, we summarize the connection between AOP networks and BNs and explore the implications for toxicity modeling. We also present a case study in drug-related liver toxicity. Our results confirm that AOP networks are congruent mathematically with BNs, with incorporation of the mathematical properties of BN leading to significantly simplified and more efficient models.

Keywords: adverse outcome pathway; Bayesian networks; drug development; *in vitro* assays; hepatotoxicity; toxicity prediction

Introduction

Chronic animal testing is still considered the gold standard of toxicity testing, in both the evaluation of environmental chemicals and drug development. However, it is now widely recognized that animal-based testing has important shortcomings. In the drug discovery process, animal testing often fails to predict the human toxicity of drug candidates. Even when toxicity can be detected, animal testing often can only provide limited insight into the biological pathways that are involved. Testing based on *in vitro* assays^{(p1),(p2)} can be directly anchored by human biology for toxicity responses; thus, the results are more relevant to human biology with clear interpretations. Partially replacing animal-based testing with *in vitro* assays is also consistent with societal demand for reducing animal use in toxicity evaluations.^{(p3),(p4)} There is a significant literature on, as well as resources for, developing molecular assays that could be used

to develop predictive models for toxicity.^{(p2),(p5),(p6),(p7),(p8),(p9)} Nevertheless, although there is substantial focus on developing machine learning models to predict toxicity,^{(p10),(p11),(p12)} how to develop accurate and efficient models suitable for practical use remains a major challenge.

The AOP framework has the potential to facilitate the development of predictive models for toxicity with *in vitro* assays. AOPs^{(p13),(p14)} and AOP networks are knowledge management frameworks for efficiently storing and accessing mechanistic knowledge required for assessing toxicity. AOPs can be utilized to aid toxicity evaluations for chemicals both qualitatively and quantitatively. For quantitative assessment, there is significant interest in combining AOPs with BNs because of their shared representation using DAGs. With many advantages of BNs in probability reasoning,^(p15) success has been reported in performing inference for toxicity by combining AOP networks with BN

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modeling.^{(p16),(p17),(p18),(p19),(p20),(p21),(p22),(p23)} However, the application of BNs as statistical models requires important mathematical assumptions. Although AOP networks and BNs share the topological structure of DAGs, whether AOP networks are mathematically congruent with BN has not been verified empirically. Furthermore, even with studies using BNs for AOP-based modeling, important properties, such as Markov blankets, have not been emphasized in the model development process, which is a missed opportunity for simplifying and optimizing the model. A Markov blanket is the minimal set of nodes that, if their states are known, renders a target node conditionally independent of all other nodes in the network. It can be considered as a cocoon separating the state of the node inside from its external environment. This property is extremely useful for making inference, as we discuss below.

Here, we first summarize the connection between AOP networks and BNs and explore the implications for toxicity modeling. We introduce the important concepts in BN (Markov blanket and d-separation), which will be beneficial for developing optimized models. We illustrate these concepts with a case study in the prediction of drug hepatotoxicity potential. Our results confirm that AOP networks are mathematically congruent with BNs, and that the incorporation of the mathematical properties of BNs leads to significantly simplified and more efficient models for the prediction of hepatotoxicity potential.

AOP and AOP networks

AOP is an analytical construct that describes a sequential chain of causally linked events at different levels of biological organization, which lead to an adverse effect at individual or population level^{(p13),(p14)} (Figure 1).^(p23) AOPs are the central element of a toxicological knowledge framework being built to support chemical risk assessment based on mechanistic reasoning. AOPs are described and constructed by considering biological plausibility and available evidence for how a perturbation of the biological system can lead, through various steps, to a specific adverse outcome (see <https://aopkb.org/> and <https://aopwiki.org/> for more details). In its basic form, an AOP is presented as a linear chain of events, from the molecular initiating event (MIE), through a sequence of key events (KEs), leading to the apical adverse outcome. For the example in Figure 1, aryl hydrocarbon receptor (AHR) activation is the MIE, which leads to CYP1A2/5 induction, which, in turn, leads to uroporphyrinogen oxidation, and so on.

Eventually, this chain of KEs concludes in uroporphyrin, the adverse outcome at the organism level. The linear construction of AOPs is perfectly suited to convey mechanistic knowledge and assist in determining the mode of action and developing *in vitro* tests.

Given that biological processes are usually very complex, multiple related AOPs inevitably share some key events. This naturally leads to AOP networks of various complexity, as illustrated by the AOP network for cholestasis, subsets of which are detailed in Figure 2a).^(p24) One important property of AOP networks is that they are represented as DAGs. In graph theory, a DAG comprises vertices (nodes) and edges (arcs), with each edge leading from one vertex to another, such that following those directions will never form a closed loop. In AOP networks, each KE might be caused by one or more preceding KEs (or MIEs) and can lead to one or more subsequent KEs, as indicated by the direction of edges or arcs (key event relationships or KERs). If one starts from a MIE and follows the direction of KERs, one will eventually arrive at the adverse outcome at the individual or population level. This representation of mechanistic knowledge with DAGs is consistent with the usual human reasoning process for causal relationships. It also provides a natural connection with the formalism of BNs, which we touch on again below.

The AOP framework has matured from a largely conceptual construct to an increasingly practical tool with many applications in risk assessment and regulatory decision-making.^{(p25),(p26),(p27)} However, a comprehensive survey of the application of the AOP framework is beyond the scope of this review. Nevertheless, before concluding this section, we highlight interesting examples of the application of AOP networks in addition to the example for liver toxicity, which is discussed in the case study.

Given that the endocrine system regulates tissue and organ development and function, endocrine disruptors (EDs) can lead to serious health effects.^{(p28),(p29)} However, because the endocrine system involves many different hormones and signaling pathways, constructing AOPs for EDs has received significant interest. Ziliacus *et al.*^(p30) assembled an AOP network for 42 AOPs related to mammalian reproductive toxicity arising from disruption to the estrogen, androgen, and steroidogenesis modalities. They further identified 16 subnetworks related to effects on hormone levels or hormone activity, cancer outcomes, male and female reproductive systems, and overall effects on fertility and

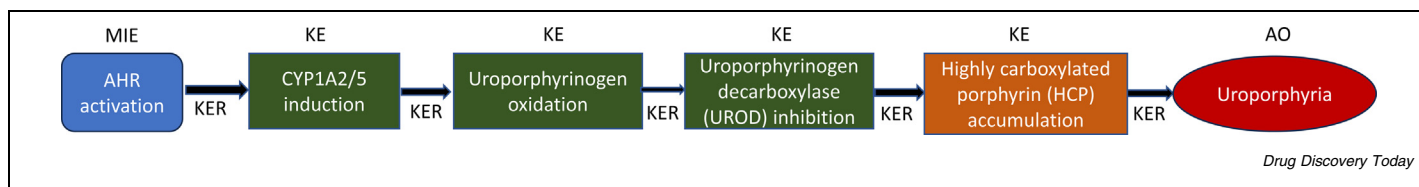
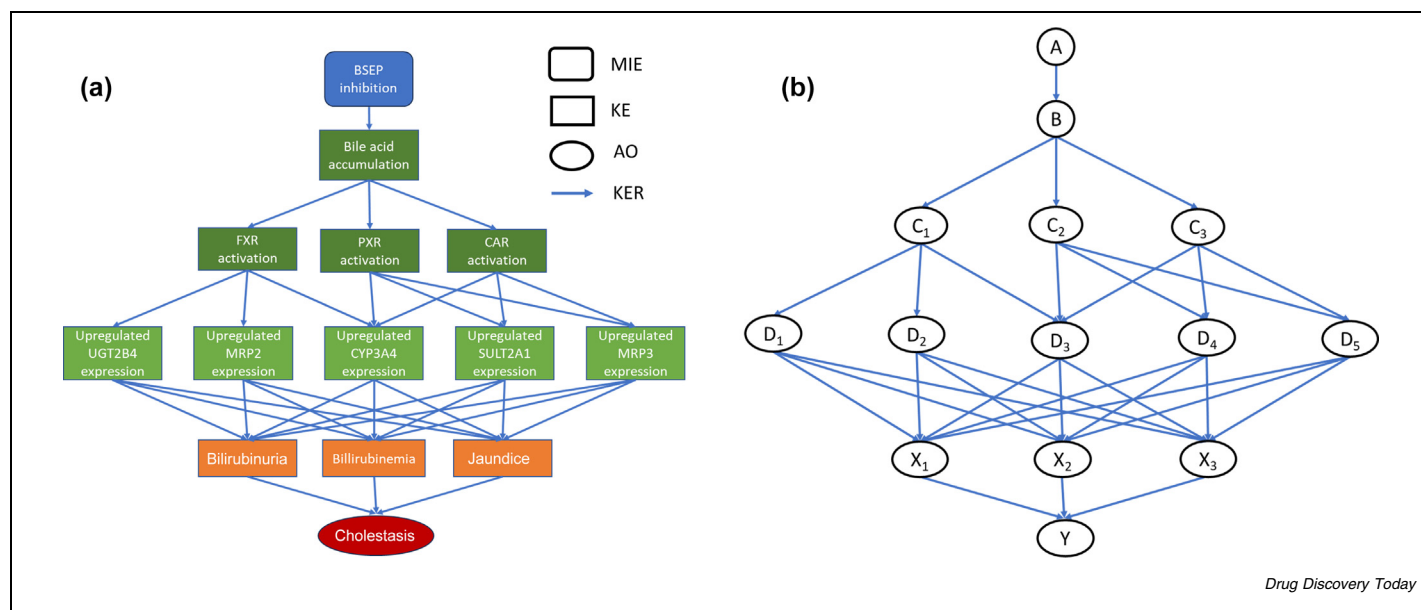


FIGURE 1

Adverse outcome pathway (AOP): aryl hydrocarbon receptor (AHR) activation leading to uroporphyrin. This AOP is listed as AOP 131 in AOPwiki. Abbreviations: AO, adverse outcome; KE, key event; KER, key event relationship; MIE, molecular initiating event.

**FIGURE 2**

An illustrational example for adverse outcome pathway (AOP) networks and the corresponding Bayesian network (BN). **(a)** Illustrational AOP network for cholestasis. The graph only contains a subset of events for cholestasis and is intended for illustration purpose only. It demonstrates the structure of a directed acyclic graph (DAG). **(b)** A BN with the same structure as the AOP network in (a). Nodes are relabeled for conciseness in later discussion in the main text. Abbreviations: AO, adverse outcome; KE, key event; KER, key event relationship; MIE, molecular initiating event.

reproduction. In total, 26 KEs and 11 KERs were identified as core blocks of knowledge in the AOP network, which could support the identification and assessment of EDs.

Drawing on data from the comparative toxicogenomics database (CTD) and AOP wiki, Chai *et al.*^(p31) constructed a AOP network for inorganic arsenic-induced adult male reproductive impairment with 16 AOPs. This provided global views of KEs in the network and highlighted the essential roles of testosterone reduction, response to reactive oxygen species, and spermatogenesis. It also helped to address existing knowledge gaps in this area for future experimental verification.

In terms of neurotoxicology, Spinu *et al.*^(p32) described an AOP network integrating nine linear AOPs. The authors identified several critical KEs, with cell injury/death established as the most hyperlinked KE across the network. The derived network enabled better understanding of signaling pathways involved in developmental and adult/ageing neurotoxicity and could provide abasis to guide the development of *in vitro* tests supporting hazard characterization and chemical risk assessment.

Finally, there have been advances in computational methods to assist the construction, analysis, and visualization of AOP networks.^{(p33),(p34),(p35)}

Bayesian networks

BNs are a class of graphical models that enable simplified representation of probabilistic dependencies between a set of random variables as a DAG. Here, we introduce the basic concepts, and the presentation is not mathematically rigorous. Detailed discussions of BNs can be found elsewhere.^{(p36),(p37)} A graph comprises a set of nodes or vertices and a set of connections between pairs of vertices called arcs, edges, or, informally, arrows. Figure 2b

details such a graph, which has the same graphical structure as the AOP network in Figure 2a, but relabeled for the clarity of our subsequent discussion. In BNs, each node of the graph corresponds to a random variable. If one follows the directions of the arcs, there should be no loops (directed acyclic graph). If two nodes are connected by an edge, depending on the direction of the edge, they form a parent–child pair. For example, in Figure 2b, node D_3 has parents C_1 , C_2 , and C_3 and has children X_1 , X_2 , and X_3 . Similarly, nodes on the same directed path are ancestors or descendants even if not directly connected by an edge. In Figure 2b, nodes A and B are ancestors of D_3 although they are not parents. Similarly, Y is a descendant of D_3 but not a child.

An essential property of BNs is the Markov property. If we index all the nodes in the DAG of BN as $\mathbf{Z} = \{Z_1, Z_2, \dots, Z_p\}$, the joint probability distribution of P_z can be written as Equation (1):

$$P_z(\mathbf{Z}) = \prod_{i=1}^p P_{Z_i}\{Z_i | PA(Z_i)\} \quad (1)$$

Here, we use $PA(Z_i)$ to designate the set of parent nodes of Z_i . This property greatly facilitates computation of probabilities regarding any subset of variables on the graph.

Given that both AOP networks and BNs share the structure of DAGs, it is natural to build BNs on AOP networks and use them to predict the adverse outcomes using the measurements regarding KEs in AOPs. AOPs suggest causal links between upstream KEs and downstream KEs or outcomes. In BNs, the joint probability distribution is written as the product of the conditional distributions of child nodes given their parent nodes (Equation (1)). If the causal relationships of AOPs are valid, it is reasonable to assume that the probability distributions of downstream KEs

depend on upstream events. Thus, modeling AOP networks with BNs is conceptually appealing. If AOP networks can be modeled with BNs with known parameters, one can perform not only inference on adverse outcomes based on the information of preceding KEs, but also ‘reverse’ inference on events given the observations on the nodes that are descendants. There are different ways to set the parameter in the BN model. Approaches in published work include using expert knowledge and learning the parameters from a curated data set. One important advantage of applying the Bayesian framework for modeling is that model uncertainty can be quantified naturally, which is vital for risk assessment applications.

In this study, we focus on the problem of predicting toxicity potential as the apical adverse outcome, that is, Y in Figure 2b. In principle, one can achieve this by computing the probability associated with Y using the entire network in Figure 2b. However, we will show that this task can be simplified substantially by leveraging some basic properties of BNs. For this, we introduce the idea of d-separation in BN, which is a criterion for determining whether two sets of variables are independent given a third set. Readers can consult relevant sources^{(p36),(p37)} for the technical discussion of d-separation. Conceptually, if a set W of nodes d-separate set U and set V in a DAG, set U and set V are statistically independent given set W . This means that, if we have information on set W , set U will provide no additional information for set V , and vice versa. Thus, we can try to find a set W that is convenient to measure for the inference of V without worrying about nodes in U . How to identify the sets for d-separation is discussed in the references given above. However, here we focus on sets that d-separate the end outcome (node Y) from some of its ancestors (e.g., node A). For this, we can simply find a set of nodes that block every directed path of arcs from node A to node Y . For example, $C = \{C_1, C_2, C_3\}$ is such a d-separation set. Similarly, $X = \{X_1, X_2, X_3\}$ is also a d-separation set. The practical implication for modeling is that, if we have measurements on nodes in C , nodes A and B will not provide any further useful information for the inference of Y . Similarly, if nodes in X were measured, all the nodes above X will not provide useful information regarding Y . This property of d-separation can be used to construct more sparse models for practical applications, as we show in our example in the next section. The set X , especially, is a Markov blanket (see references above) regarding Y , which is the minimal set of nodes that provides all the useful information for Y for all other nodes that are above it.

Although AOP networks and BNs share the underlying structure of DAG, whether AOP networks are in fact consistent with BNs and have the Markov property is an assumption that one needs to make for the model. In the next session, we show that, for our example, the result based on the liver toxicity AOP network is congruent with the mathematical property of BNs.

Finally, we want to comment on the computational aspects of BN models. When a BN is large, making inference involves computation over a large number of nodes, which could be computationally intensive. Efficient algorithms exist when all the nodes represent discrete random variables or continuous variables following a normal distribution.^(p37) Thus, many software packages

only allow nodes to be discrete or normally distributed. This sometimes leads to the misconception that BNs can only be applied to discrete or normal random variables. In fact, random variables in BNs can take a range of probability distributions. However, with more general distributions, exact calculation of probabilities often becomes impractical, and Markov chain Monte Carlo (MCMC) simulation becomes the preferred method for implementation. This can be carried out with general purpose Bayesian modeling packages, such as JAGS.^{(p38),(p39),(p40)} In our example, we use JAGS to implement a sparse Bayesian model that leads to superb performance.

Prediction for drug liver toxicity, a case study

Here, we use drug liver toxicity to illustrate how to utilize the properties of BNs to facilitate model building in the prediction of adverse outcomes. It also provides an opportunity to empirically evaluate whether model presentation of AOP networks is congruent with the mathematical properties of BNs. Drug-related liver toxicity is one of the major safety concerns in drug development and approval.^{(p41),(p42)} Accordingly, there has been significant effort to describe AOPs related to liver toxicity.^{(p43),(p44),(p45),(p46),(p47),(p48)}

AOP networks for liver toxicity

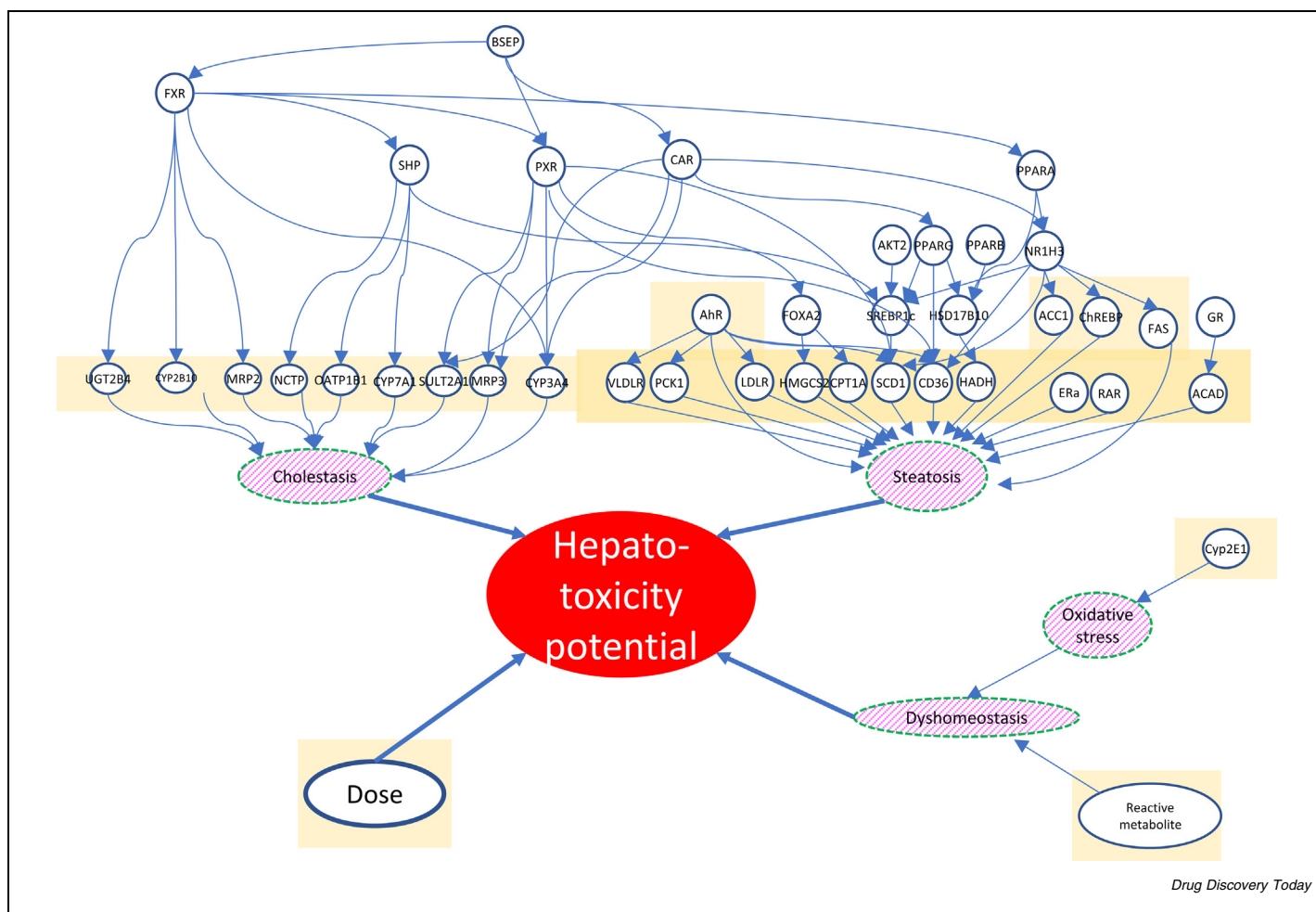
We built an AOP network for modeling drug liver toxicity (Figure 3), using liver-related AOPs collected in AOPwiki (<https://aopwiki.org/>) and from the literature.^{(p24),(p43),(p49),(p50)} Drug-related liver toxicity involves many biological processes and the corresponding AOP networks can be presented in multiple ways depending on the chosen granularity. Thus, we built AOP networks with published AOPs emphasizing molecular events that have available *in vitro* assays to illustrate BN modeling; it is not meant to be an exhaustive examination of mechanisms of drug-related liver toxicity. The pink ovals in Figure 3 are used to group different AOPs for clarity and are not used in the model. For conciseness, Figure 3 only included molecular events for which we had obtained data. There are numerous other KEs that could be included in the model, such as bilirubinuria and triglyceride accumulation. We omitted most of these KEs except those represented by pink ovals in Figure 3 because they were not used in the BN model or to keep the graph at a reasonable size.

BN models

Given that the random variables associated with gene expression are continuous, we assumed that they had a normal distribution, with the mean being a linear function of its parent nodes. For the response, hepatotoxicity, which is a binary variable, we assumed a logistic model (Equation (2)):

$$\log \frac{P(Y = 1 | \mathbf{X} = \mathbf{x})}{P(Y = 0 | \mathbf{X} = \mathbf{x})} = \beta_0 + \boldsymbol{\beta}^T \mathbf{x}, \quad (2)$$

where Y is the hepatotoxicity potential (0 means no, 1 means yes) and the vector $\mathbf{X}$ is the value on all the parent nodes of Y . In Figure 3, the parent nodes of Y are the 27 nodes in yellow. In total, the BN associated with AOP network in Figure 3 contains 42 nodes, including the response, hepatotoxicity potential (Y). As mentioned earlier, the pink ovals (cholestasis, steatosis, dyshome-

**FIGURE 3**

An adverse outcome pathway (AOP) network for drug hepatotoxicity potential. The pink ovals are presented for narrative clarity and are not included in the model. Nodes in the yellow area form the Markov blanket of hepatotoxicity potential. We only included molecular events with available data in this graph; it is not meant to be an exhaustive presentation of liver toxicity-related AOPs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ostasis, and oxidative stress) are presented solely for mechanistic clarity and are not part of the BN model.

Following the concepts outlined in the previous section, the parent nodes of Y form a Markov blanket for hepatotoxicity potential. When we have values for these 27 variables, other nodes do not provide additional information for predicting the state of Y . We can infer the probability of the presence of hepatotoxicity potential, $P(Y = 1)$, with the model in Equation (2). To fully specify the Bayesian model, we adopt horseshoe priors^{(p55), (p56), (p57)} for each regression coefficient in $\beta = \{\beta_k\}_{k=1}^p$. We assume that β_k has a prior distribution of scale mixture of normal (Equation (3)):

$$\beta_k | \gamma_k, \tau_k \sim N(0, \gamma_k^2 \tau_k^2), \gamma_k \sim C^+(0, 1) \quad (3)$$

where γ_k is a local shrinkage parameter, τ is an over-all global shrinkage parameter, and C^+ is the half-Cauchy distribution. Carvahlo and coworkers^(p58) showed that, for variables representing pure noise without useful information, the horseshoe priors can induce shrinkage of their coefficients to zero. For variables with a real effect on the response, the shrinkage effect on coefficients

by the horseshoe prior is mild. Our experience showed that adoption of the horseshoe prior in this setting was advantageous. Without proper shrinkage of the coefficients, the model can become unstable with inferior performance. We used JAGS^(p38) and the *Rjags* package (<https://cran.r-project.org/package=rjags>) to perform inference for this Bayesian model with MCMC. Application of the horseshoe prior was unfeasible if we limited the types of distribution to be either discrete or normal. Thus, using a generic Bayesian modeling software such as JAGS is essential to fit more flexible models for BN.

Results and comparisons for different models

We fitted several BN models based on the AOP network in Figure 3 (or its subsets) to obtain their parameters with the data described above. The probability for the drug to have hepatotoxicity potential was then computed with the BN; drugs with probability >0.5 based on molecular assays were considered to have hepatotoxicity potential. We compared the result with the classification of DILI score^(p51) to calculate sensitivity, specificity, and accuracy. As a consequence of Equation (1) and the Markov

TABLE 1

Results for several BN models for drug hepatotoxicity potential

Model	True positive	False positive	True negative	False negative	Sensitivity	Specificity	Accuracy
Markov blanket/Full	84	9	45	8	0.903	0.849	0.884
Flat	84	9	45	8	0.903	0.849	0.884
Upper set	83	12	42	9	0.874	0.823	0.856
Upper partial set	83	22	32	9	0.790	0.780	0.788

property for BN as discussed previously, the probability distribution of the hepatotoxicity potential, Y , was entirely determined by the values of variables in the Markov blanket. Thus, the BN based on the entire network in Figure 3 (41 nodes plus the response) gave the same result as the BN based only on the nodes in the Markov blanket (the 27 nodes in the yellow shade) if we were only interested in making predictions for the hepatotoxicity potential. Thus, we show the results in a single row (Markov blanket/Full) in Table 1. They gave a sensitivity of 0.903, specificity of 0.849, and accuracy of 0.884 (Table 1).

If AOP networks can be reasonably modeled as BNs, it means we only need to use the subset of nodes (27 nodes) in the Markov blanket for modeling rather than all the nodes (41), which makes the computation burden much smaller and requires fewer assays. However, this is an assumption that has not been empirically evaluated. To assess whether AOP networks are congruent with the mathematical properties of BN, we also fitted a flat' (Figure 4a). The flat model is a logistic model where every *in vitro* assay can affect the risk of liver toxicity with the horse-shoe priors on each regression coefficient. If the Markov property does not hold for the AOP network in Figure 3 and nodes outside of the Markov blanket can provide useful information for the hepatotoxicity potential, the flat model should perform better than the Markov blanket-based model or the full BN model. Results in Table 1 (first two rows) shows that the performances were, in fact, the same, confirming that AOP networks are congruent with the Markov property assumption.

With the Markov property, the BN model using only nodes in the Markov blanket fully utilizes all the information contained in the AOP network. However, it might sometimes be preferable to find a subset that has an even smaller number of nodes than the Markov blanket. This is useful for reducing the cost and logistic burden for performing *in vitro* assays at the expense of reduced model performance because of the lost information. For this, we fitted the 'upper set' model with nodes in the turquoise region in Figure 4b. This set included nine nodes, fewer than the 27 for the Markov blanket. One property of this set of nodes is that they formed a d-separation set between the response (hepatotoxicity potential) and the MIEs; that is, all the directed paths from the MIEs to the response had to pass through one of the nodes in this set. Table 1 (Row 3) shows that this upper set model had reduced performance compared with that based on nodes in the Markov blanket (sensitivity 0.874, specificity 0.823, and accuracy 0.856), although the differences are not dramatic.

We also explored a model that we called the 'upper partial set' model. The upper partial set included the nodes in the upper set except the nodes FXR and reactive metabolite; thus, this set had seven nodes. Here, FXR is the ancestor of several genes down-

stream and reactive metabolite is a drug property. Removing these two nodes broke down the d-separation between the initiating events and the final response because there were paths from the initiating events to the response through the two excluded nodes. Table 1 (Row 4) shows that the performance of this model deteriorated more significantly (sensitivity 0.790, specificity 0.780, and accuracy 0.788), especially for specificity.

Discussion

AOP is a valuable framework for knowledge organization related to toxicity risk assessment. Given that AOP networks share the underlying DAG structure with BNs, it is natural to perform inference tasks by combining these two concepts. Using AOP-based BNs for risk assessment has shown some success.^{(p16),(p17),(p18),(p19),(p20),(p21),(p22),(p23)} However, as statistical models, with specific assumptions and mathematical properties, BNs have not been carefully evaluated in the context of AOP networks. Thus, we used liver toxicity AOP networks to empirically assess whether the Markov property holds for AOP networks. Our results are congruent with the nodes in the Markov blanket containing all the information to predict the response (hepatotoxicity potential), which supports the use of BNs for AOP based modeling. This also suggests that one can prioritize nodes in the Markov blanket for prediction, thus significantly reducing the number of *in vitro* assays that are required. Burgoon *et al.*^(p16) presented an approach based on the backdoor algorithm (Pearl1993)^(p59) with the same effect. The concepts of Markov blanket and d-separation that we present here provide a more convenient framework to identify a subset of nodes (assays).

The structure of AOP networks also provides additional information that can be utilized to facilitate model building and assay development. We explored using a subset of nodes other than the Markov blanket. The upper set of nodes had fewer nodes (nine versus 27), but it still formed d-separation between the initiating events and the response. The model using the upper set had a moderately decreased performance compared with that using nodes in the Markov blanket. When the d-separation property was violated, such as with the upper partial set, the deterioration of performance became more severe. This points to an approach to select a smaller set of assays to reduce cost while accepting a mild reduction in model performance. As one reviewer pointed out, nodes 'upstream' in an AOP typically represent lower levels of biological organization (cells or tissues) and assays performed at lower levels of biological organization can be less costly in both money and time. It might also reduce the use of animal for drug testing. These are additional arguments to support the idea that a smaller set of assays closer to the MIE might be optimal. When feasible, a set of nodes that cause d-separation

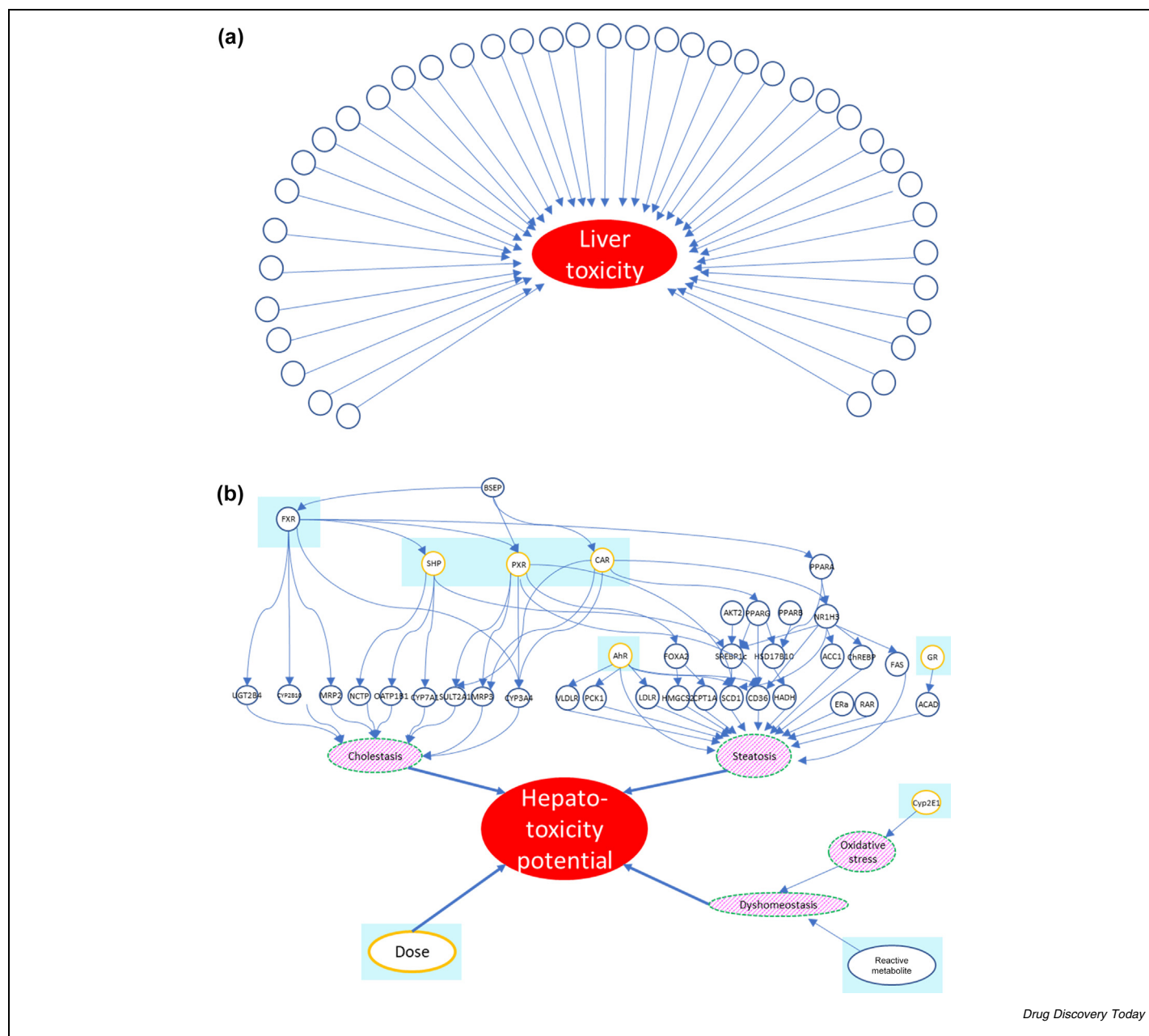
**FIGURE 4**

Illustration of the flat model and the partial upper set model. **(a)** The Bayesian network (BN) with the flat model. All 41 variables can provide information for the prediction of liver toxicity risk. **(b)** Illustration for the upper set. Nodes in the upper set are in the turquoise area.

between the initiating events and the response tends to enhance model performances.

In our Bayesian model for liver toxicity, we adopted horseshoe priors for the coefficients, which was crucial for good model performance. This suggests the advantage of using a broad range of distribution types in BNs rather than being limited to discrete and normal distributions. To allow a broad range of distributions, a generic Bayesian modeling software is required, such as JAGS used in our example. In the Bayesian modeling framework, even more sophisticated models can be considered. For example, we assumed a linear relationship for the mean of gene expression levels in specifying the KE node given its parent nodes. This model gave satisfactory results in the case study. However, nonlinear relationships or

nonparametric forms could be implemented with modern Bayesian statistical software. As in any modeling exercise, one needs to balance the power of more flexible models with the cost of model complexity, an aspect that could be explored further in future studies. Our example also demonstrated that there is potential in combining AOP and BN in the drug development process. Given that AOPs are being continuously developed and BN is a highly powerful statistical approach, more advances that combine these two approaches are expected.

Finally, as with any model, BNs involve certain levels of abstraction. In our discussions on the connection between AOP networks and BNs, some of the properties will strictly hold only if our knowledge is perfect, that is, the AOP networks encapsulate

all the possible pathways leading to the adverse outcome. This assumption might not always be true in practice, given the enormous complexity of biology. When high-resolution data become available, it will be possible to discover new pathways and relationships leading from upstream nodes to the AOP. Thus, models should evolve with new knowledge in biology, similar to the continuing evolution of AOPs.

Author contributions

D.W.: writing – review & editing; writing – original draft; investigation; data curation; conceptualization. **A.S.:** writing – review & editing; writing – original draft; validation; conceptualization. **W.T.:** writing – review & editing; conceptualization.

Data availability

Data for hepatotoxicity classification are available through Liver Toxicity Knowledge Base (<https://www.fda.gov/science-research/bioinformatics-tools/liver-toxicity-knowledge-base-ltkb>). Gene expression data from the L1000 project can be obtained from <https://lincsproject.org/LINCS/tools/workflows/find-the-best-place-to-obtain-the-lincs-l1000-data>.

Data

As the response variable of the BN model with the DAG in Figure 3, drug hepatotoxicity potential was obtained from the Liver Toxicity Knowledge Base (LTKB).^(p51) LTKB is a project of the US Food and Drug Administration (FDA)'s National Center for Toxicological Research that focuses on drug-induced liver injury (DILI), which provides several benchmark data sets regarding the liver toxicity of drugs. Chen and coworkers^(p51) implemented the DILI score, a classification algorithm of hepatotoxic potential for 354 drugs (124 most concern, 162 less concern, and 68 no concern) based

on FDA labels, DILI case registries, publications, and other sources. In our model, we used most concern (drugs with hepatotoxic potential or case drugs, $N = 92$) and no concern (drugs with no evident hepatotoxic potential or control drugs, $N = 54$) drugs in LTKB for which molecular assay information was available. For random variables associated with molecular events in the AOP network, we use normalized expression change for the genes under treatment with each drug. Data for gene expression levels from L1000 assays^(p9) were obtained from the Gene Expression Omnibus with the accession numbers GSE92742 and GSE70138. L1000 is a high-throughput bead-based gene expression profiling assay that can be used to efficiently generate whole-transcriptome profiles for cells under various treatments. Data for L1000 assays were provided in several different formats. For our analysis, the replicate-collapsed z-scores (Level 5 data) were extracted with the CMapR package^(p52) and further processed as described previously.^(p53) The status of reactive metabolite formation and dose was obtained from LTKB.^(p51) Here, dose was the only random variable in the model that was not a molecular event or adverse outcome in the AOP network. Previous work has shown that dose is a critical variable in predicting drug liver toxicity.^{(p51),(p54)}

Data availability

Data will be made available on request.

Acknowledgment

The opinions expressed in this paper are those of the authors, and do not necessarily reflect the position of the FDA.

Conflicts of interest

The authors declare no conflicts of interest.

References

- Kavlock R, Dix D. Computational toxicology as implemented by the US EPA: providing high throughput decision support tools for screening and assessing chemical exposure, hazard and risk. *J Toxicol Env Health, Part B*. 2010;13:197–217.
- Tice RR, Austin CP, Kavlock RJ, Bucher JR. Improving the human hazard characterization of chemicals: a Tox21 update. *Environ Health Persp*. 2013;121:756–765.
- Rudén C, Hansson SO. Registration, evaluation, and authorization of chemicals (REACH) is but the first step—how far will it take us? Six further steps to improve the European chemicals legislation. *Environ Health Persp*. 2010;118:6–10.
- Locke PA et al. Implementing toxicity testing in the 21st century: challenges and opportunities. *Int J Risk Assess Manage*. 2017;20:198–225.
- Collins FS, Gray GM, Bucher JR. Transforming environmental health protection. *Science*. 2008;319:906–907.
- Judson R et al. In vitro and modelling approaches to risk assessment from the US Environmental Protection Agency ToxCast programme. *Basic Clin Pharmacol*. 2014;115:69–76.
- Richard AM et al. ToxCast chemical landscape: paving the road to 21st century toxicology. *Chem Res Toxicol*. 2016;29:1225–1251.
- Igarashi Y et al. Open TG-GATEs: a large-scale toxicogenomics database. *Nucleic Acids Res*. 2015;43:D921–D927.
- Subramanian A et al. A next generation connectivity map: L1000 platform and the first 1,000,000 profiles. *Cell*. 2017;171:1437–1452.
- Liu J et al. Predicting hepatotoxicity using ToxCast in vitro bioactivity and chemical structure. *Chem Res Toxicol*. 2015;28:738–751.
- Huang R et al. Modelling the Tox21 10 K chemical profiles for in vivo toxicity prediction and mechanism characterization. *Nat Commun*. 2016;7:10425.
- Kleinstreuer NC et al. Development and validation of a computational model for androgen receptor activity. *Chem Res Toxicol*. 2017;30:946–964.
- Ankley GT et al. Adverse outcome pathways: a conceptual framework to support ecotoxicology research and risk assessment. *Environ Toxicol Chem*. 2010;29:730–741.
- Krewski D et al. Toxicity testing in the 21st century: a vision and a strategy. *J Toxicol Env Health, Part B*. 2010;13:51–138.
- Pearl J. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. San Mateo: Kaufman; 1988.
- Burgoon LD et al. Predicting the probability that a chemical causes steatosis using adverse outcome pathway Bayesian networks (AOPBNs). *Risk Anal*. 2020;40:512–523.
- Gust KA et al. Limitations of toxicity characterization in life cycle assessment: can adverse outcome pathways provide a new foundation? *Integr Environ Assess Manag*. 2015;12(3):580–590.
- Jeong J, Choi J. Quantitative adverse outcome pathway (qAOP) using bayesian network model on comparative toxicity of multi-walled carbon nanotubes (MWCNTs): safe-by-design approach. *Nanotoxicology*. 2022;16:679–694.
- Mentzel S et al. Evaluating the effects of climate change and chemical, physical, and biological stressors on nearshore coral reefs: a case study in the Great Barrier Reef. *Australia Integr Environ Assess Manag*. 2024;20:401–418.
- Moe SJ, Carriger JF, Glendell M. Increased use of Bayesian network models has improved environmental risk assessments. *Integr Environ Assess Manag*. 2020;17:53–61.
- Moe SJ et al. Quantification of an adverse outcome pathway network by Bayesian regression and Bayesian network modeling. *Integr Environ Assess Manag*. 2020;17:147–164.

22. Pirone JR, Smith M, Kleinstreuer NC, Burns TA, Strickland J, Dancik Y, et al. Open source software implementation of an integrated testing strategy for skin sensitization potency based on a Bayesian network. *ALTEX*. 2014;31:336–340.
23. AOP: Aryl hydrocarbon receptor activation leading to uroporphyrin. AOPwiki website. <https://aopwiki.org/aops/131> [Accessed March 30, 2025].
24. Vinken M. The adverse outcome pathway concept: a pragmatic tool in toxicology. *Toxicol*. 2013;312:158–165.
25. Ankley GT, Edwards SW. The adverse outcome pathway: a multifaceted framework supporting 21st century toxicology. *Curr Opin Toxicol*. 2018;9:1–7.
26. Street ME, Audouze K, Legler J, Sone H, Palanza P. Endocrine disrupting chemicals: current understanding, new testing strategies and future research needs. *Int J Mol Sci*. 2021;22:933.
27. Bajard L et al. Application of AOPs to assist regulatory assessment of chemical risks—case studies, needs and recommendations. *Environ Res*. 2023;217, 114650.
28. Gore AC et al. EDC-2: The Endocrine Society's second scientific statement on endocrine-disrupting chemicals. *Endocr Rev*. 2015;36:E1–E150.
29. Duh-Leong C, Maffini MV, Kassotis CD, Vandenberg LN, Trasande L. The regulation of endocrine-disrupting chemicals to minimize their impact on health. *Nat Rev Endocrinol*. 2023;19:600–614.
30. Zilliaceus J, Draskau MK, Johansson HK, Svingen T, Beronius A. Building an adverse outcome pathway network for estrogen-, androgen- and steroidogenesis-mediated reproductive toxicity. *Front Toxicol*. 2024;6, 1357717.
31. Chai Z et al. Generating adverse outcome pathway (AOP) of inorganic arsenic-induced adult male reproductive impairment via integration of phenotypic analysis in comparative toxicogenomics database (CTD) and AOP wiki. *Toxicol Appl Pharmacol*. 2021;411, 115370.
32. Spinu N et al. Development and analysis of an adverse outcome pathway network for human neurotoxicity. *Arch Toxicol*. 2019;93:2759–2772.
33. Yazar N et al. AOP-networkFinder—a versatile tool for the reconstruction and visualization of adverse outcome pathway networks from AOP-Wiki. *Bioinform Adv*. 2025;5, vbaf007.
34. Pittman ME, Edwards SW, Ives C, Mortensen HM. AOP-DB: a database resource for the exploration of adverse outcome pathways through integrated association networks. *Toxicol Appl Pharmacol*. 2018;343:71–83.
35. Aguayo-Orozco A et al. sAOP: linking chemical stressors to adverse outcomes pathway networks. *Bioinform*. 2019;35(24):5391–5392.
36. Neapolitan RE. *Learning Bayesian Networks*. Upper Saddle River: Pearson Prentice Hall; 2004.
37. Nagarajan R, Scutari M, Lebre S. *Bayesian Networks in R*. New York: Springer; 2013.
38. Plummer M. JAGS: A program for analysis of Bayesian graphical models using Gibbs sampling. In: Hornik K, Leisch F, Zeileis A, eds. *Proceedings of the 3rd International Workshop on Distributed Statistical Computing (DSC 2003)*. Vienna: R Foundation; 2003.
39. Carpenter B et al. Stan: a probabilistic programming language. *J Stat Softw*. 2017;76:1–32.
40. Conrady S, Jouffe L. *Bayesian Networks and BayesiaLab: A Practical Introduction for Researchers*. Franklin, Bayesia S.A.S.; 2015.
41. Chen M et al. The liver toxicity knowledge base: a systems approach to a complex end point. *Clin Pharm Therap*. 2013;93:409–412.
42. Thakkar S et al. The Liver Toxicity Knowledge Base (LKTb) and drug-induced liver injury (DILI) classification for assessment of human liver injury. *Expert Rev Gastroenterol Hepatol*. 2018;12:31–38.
43. Gijbels E, Vinken M. An update on adverse outcome pathways leading to liver injury. *Appl Vitro Toxicol*. 2017;3:283–325.
44. Luckert C et al. Adverse outcome pathway-driven analysis of liver steatosis in vitro: a case study with cyproconazole. *Chem Res Toxicol*. 2018;31:784–798.
45. Sanz-Serrano J et al. Chemical-induced liver cancer: an adverse outcome pathway perspective. *Expert Opin Drug Saf*. 2024;23:425–438.
46. Jin Y et al. A toxicity pathway-oriented approach to develop adverse outcome pathway: AHR activation as a case study. *Environ Pollut*. 2021;268, 115733.
47. Arnesdotter E et al. Derivation, characterisation and analysis of an adverse outcome pathway network for human hepatotoxicity. *Toxicol*. 2021;459, 152856.
48. Vinken M. Adverse outcome pathway networks as the basis for the development of new approach methodologies: liver toxicity as a case study. *Curr Opin Toxicol*. 2024;40, 100504.
49. Mellor CL, Steinmetz FP, Cronin MT. The identification of nuclear receptors associated with hepatic steatosis to develop and extend adverse outcome pathways. *Crit Rev Toxicol*. 2016;46:138–152.
50. Horvat T et al. Adverse outcome pathway development from protein alkylation to liver fibrosis. *Arch Toxicol*. 2017;91:1523–1543.
51. Chen M, Borlak J, Tong W. A model to predict severity of drug-induced liver injury in humans. *Hepatology*. 2016;64:931–940.
52. Enache OM et al. The GCTx format and cmap {Py, R, M, J} packages: resources for optimized storage and integrated traversal of annotated dense matrices. *Bioinform*. 2019;35(8):1427–1429.
53. Khadka KK, Chen M, Liu Z, Tong W, Wang D. Integrating adverse outcome pathways (AOPs) and high throughput in vitro assays for better risk evaluations, a study with drug-induced liver injury (DILI). *ALTEX*. 2020;37:187–196.
54. Chen M et al. A testing strategy to predict risk for drug-induced liver injury in humans using high-content screen assays and the 'rule-of-two' model. *Arch Toxicol*. 2014;88:1439–1449.
55. Datta J, Ghosh J. Asymptotic properties of Bayes risk for the horseshoe prior. *Bayesian Anal*. 2013;8:111–132.
56. Van Der Pas SL, Kleijn BJ, Van Der Vaart AW. The horseshoe estimator: posterior concentration around nearly black vectors. *Electron J Stat*. 2014;8:2585–2618.
57. Qi X, Zhou S, Wang Y, Peterson C. Bayesian sparse modeling to identify high-risk subgroups in meta-analysis of safety data. *Res Synth Methods*. 2022;13:807–820.
58. Carvalho CM, Polson NG, Scott JG. Handling sparsity via the horseshoe. *Proc Machine Learn Res*. 2009;5:73–80.
59. Pearl J. Comment: Graphical models, causality and intervention. *Statist Sci*. 1993;8:266–269.